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Supervised Machine Learning for Liquid Rate Estimation in Oil Field with Limited Well Monitoring Data: A Case Study of the Handil Oil Field

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Abstract

Handil is an oil field located in the Mahakam swamp area. Approximately 80% of its wells are operated with gas lift. Well surveillance primarily relies on pressure, temperature, and gas injection measurements, which are locally updated through daily operator visits. In some wells, an additional single-phase orifice meter is installed on the flow line to estimate gas rates, but it doesn't provide any information on liquid rates. Mobile testing units, mounted on floating barges, conduct well tests on a campaign basis every 3 to 6 months. However, daily liquid production monitoring becomes extremely challenging due to the lack of metering references at the well level. Furthermore, fast sedimentation area restricts the mobile testing unit to approach the well cluster which further extend the well potential's uncertainties.

Supervised machine learning techniques were employed to estimate the liquid rates of the wells based on available surface parameters. Two machine learning models were developed: liquid rate prediction based on wellhead parameter data (well head estimate) and gas rate correction on water-out gas well based on orifice parameter data (orifice estimate). The preprocessed datasets consisted of clean 930 well test data for the well head estimate and 167 data for the orifice estimate. They were consisted of actual well production test data (wellhead pressure, temperature, gas injection rate, real gas rate, liquid rate, and water-cut), additional flow line parameters for the orifice estimate (flow line pressure, temperature, orifice differential pressure, beta ratio), and the open reservoir depth. The datasets were used to train and test different regression methods, including linear regression, polynomial regression, support vector machines (SVM), and random forest. Random forest regression, providing accuracy up to 93%, was selected on both ML model. Furthermore, additional polynomial regression was applied for model reinforcement, avoiding over-fit and flat prediction on the blind data beyond the training set range.

The results show that the developed machine learnings increase the field daily liquid rate production allocation accuracy at 90% from previously only 80% compared to real produced liquid, better detect liquid production decline or incline following dynamic well parameter for quick necessary action, provide new insight to sand production avoidance on specific sand control type, and accurately provide gas correction rate on water-out gas well.

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This data analytic approach for liquid rate estimation through machine learning contributes to the portfolio of techniques used in the oil and gas industry. It is highly applicable and replicable in oil fields with traditional and limited well monitoring data.

Background

Handil Field Overview

Handil, its development was started on 1975, is one of oil field located at Mahakam delta rivers. Some of wells are gas producers, but 80% of the wells are oil producer operated with gas lift as artificial lift means. The field own produced gas is compressed and re-circulated back as well's gas lift source.

Handil's wells are grouped into 66 clusters based on their surface location illustrated on figure 1. By its design, there are no production testing facilities available on each cluster. Separators are available for mainly separation purpose on the process area, while coriolis flow meter is located on the tail of the processing area to measure bulk oil rate of the field. Handil process flow is illustrated on figure 2.

Well surveillances are limited to wellhead flowing pressure & temperature reading (WHFP, WHFT) and gas lift injection rate calculated from gas lift flow orifice. Without any real time monitoring device, these parameters are locally updated through operator visit once per day. There is no multiphase liquid measuring device on the well level. The surveillance relies on traditional "eyes". In some new wells installation, especially on gas well, additional single phase orifice is installed on the well flow line.

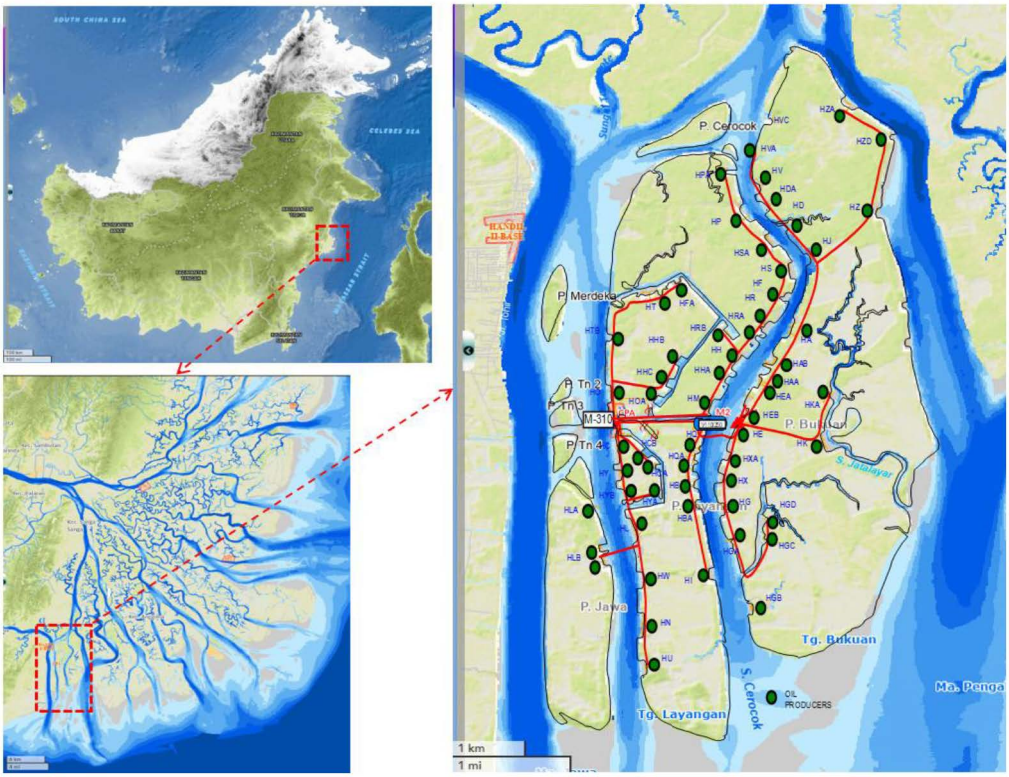


Figure 1—Field location and cluster's overview

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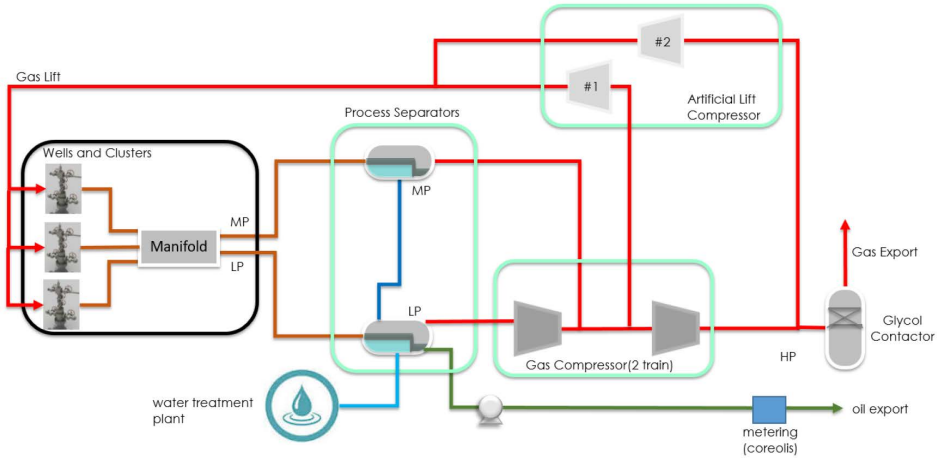


Figure 2—Simplified field flow diagram

Each cluster is equipped with two production line selection modes: low pressure and medium pressure. This configuration can be adjusted from the production manifold on each cluster. Each well can be configured and operated in either the low-pressure (LP) or medium-pressure (MP) mode. Additionally, there is a testing header available for regular well production testing using a mobile testing unit. cluster's production manifold is illustrated on figure 3.

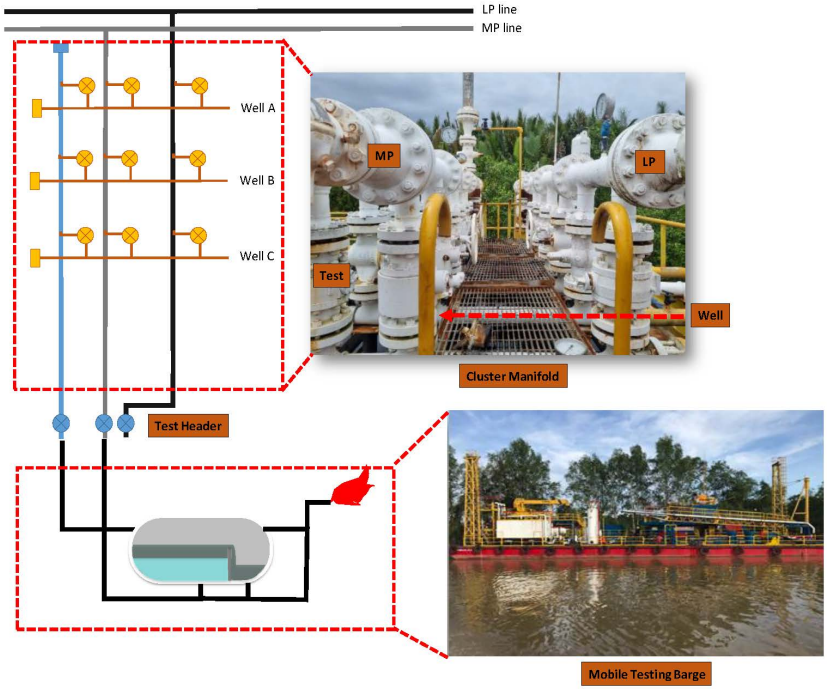
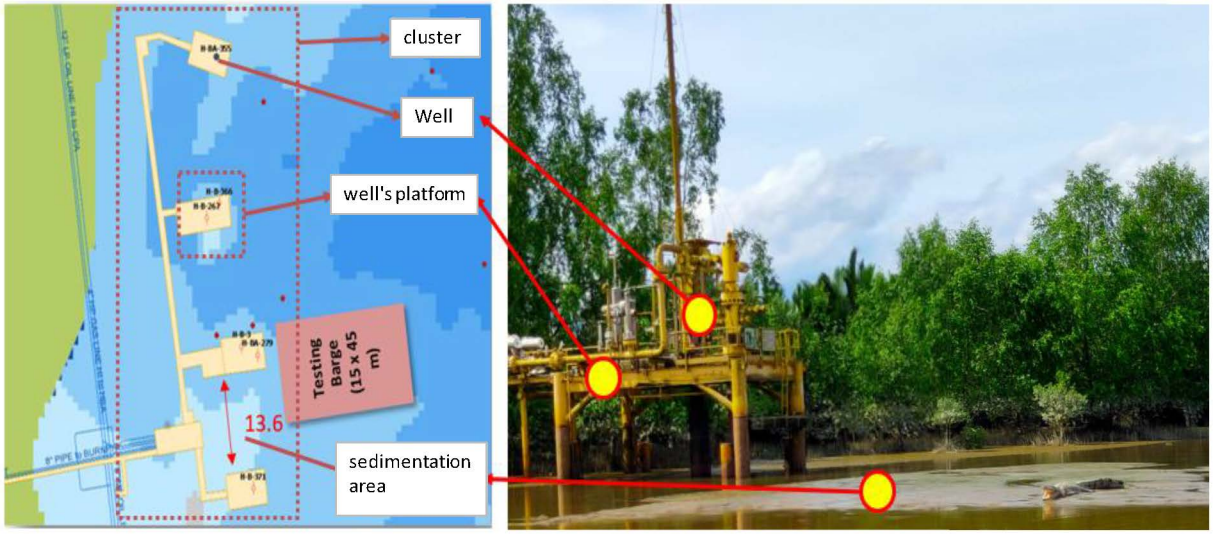


Figure 3—Simplified cluster's production manifold and mobile testing barge flow diagram

Due to its geographical condition, the testing unit is mounted on floating barge and towed to each cluster. testing equipment is lined up to cluster manifold to perform regular well testing without shut in the well in campaign basis every 4 to 6 months, ideally updated in monthly basis, due to testing barge availability. Testing barge is shown in figure 3. The main challenges of working on this environment is the fast sedimentation which restrict the barge to approach manifold as illustrated on figure 4. The water depth is insufficient to accommodate barge draft. As impact, production test couldn't be performed on the manifold resulting obsolete well production potential. directly connecting the mobile testing unit on the well

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is the other alternative. However, this method is not recommended as it requires to shut in well for testing equipment rig up purpose and well ramp up period back to previous parameter. In addition, both well and production manifold are inaccessible by testing barge in some fast sedimentation area.



(a) contour map from one of well's cluster

(b) condition on one of well's platform having high sedimentation issue

Figure 4—Illustration of high sedimentation area in one of well's cluster

Surveillance Challenges

Several challenges rise triggered by limited information on the latest well's liquid production. Highlighted issues are related to production allocation:

Network optimization. As mentioned earlier, Handil has two production mode on the well level, which are low pressure and medium pressure. In general, low pressure mode is more preferred. However, some well's production mode shall be switched to medium pressure at some specific case.

One example is the process's LP separator capacity. LP separator was designed at maximum 120 MMscfd capacity for effective separation. Once this value exceeded, some wells shall be switched to medium pressure network with impact of higher back pressure to the wells.

Well switching selection shall consider updated liquid production on specific top node pressure. Later, they will be simulated on network modeling software to ensure optimum production delivery on each well at different production mode.

Decline/ incline identification. Identifying well's rate decline/ incline is one of surveillance challenge with absence of well liquid rate measuring equipment. It will impact late production recovery or rate adjustment action. Figure 5 shows one example case where well's parameter fluctuated but the production allocation was maintained at similar value.

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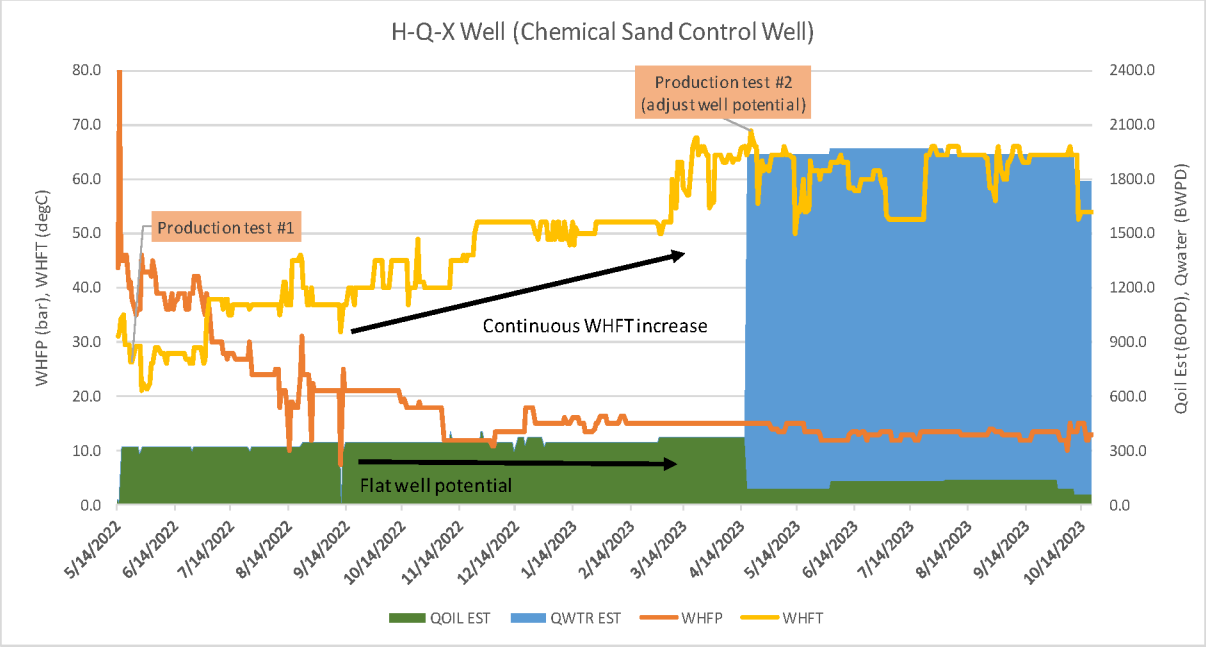


Figure 5—Unidentified well incline or decline sample case

Unmeasurable optimization led to sand production event. There is one good example case on H-X-X. This well is completed with mechanical sand control (screen) which has strict limitation on liquid production rate. Well head temperature (WHFT) was observed higher compared to clean up result without knowing the real liquid rate. Several movements were performed by increasing choke opening and gas lift rate due to WHFT drop indication. But then, well was shut in due to high sand production issue which later confirmed form slickline operation. There was no clue of real well's liquid production rate which then allocated as flat value based on clean up result. Figure 6 describes the well's production history before sand production observed.

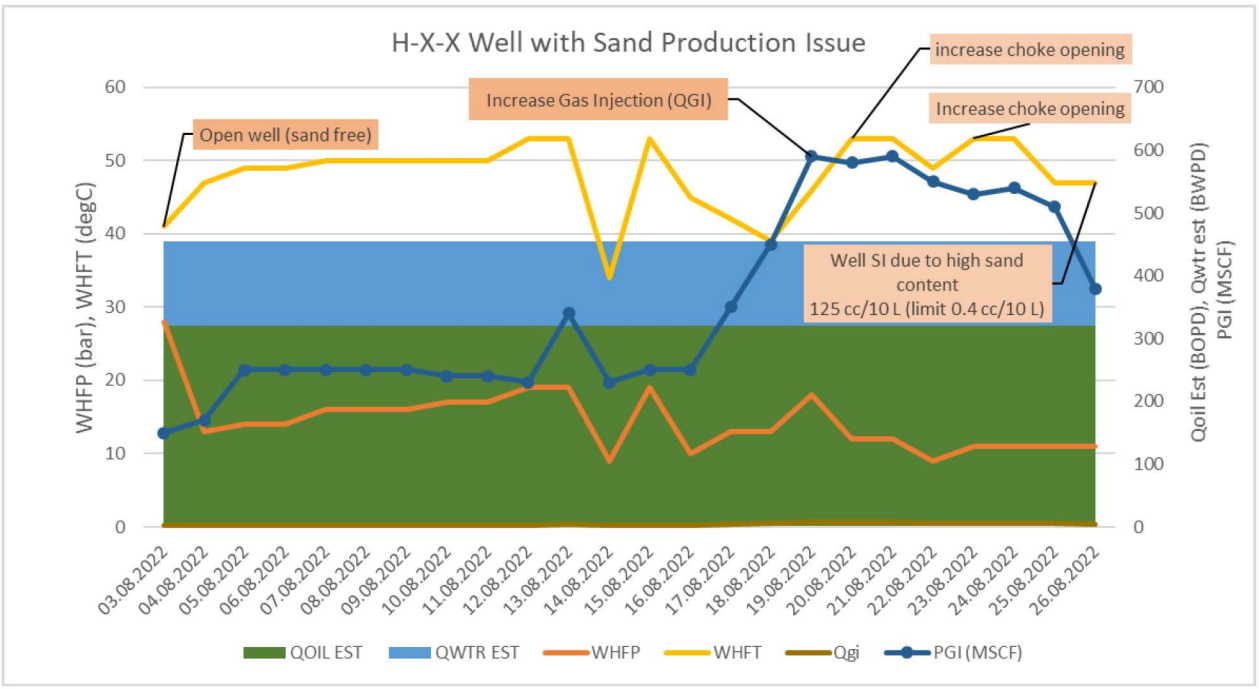


Figure 6—Unmeasurable well optimization sample case

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Well Restart priority. Any planned or unplanned shut down requires to restart all the well. The sequence will be started from the gas well to feed gas lift compressor, then continue with oil well with highest gas lift efficiency first to re-establish the production. High gas lift efficiency is defined as the well which could provide highest oil rate with similar amount of gas lift injection (BOPD/ MMscfd). Well oil rate is related with known water cut and liquid rate. Water cut may be regularly updated from liquid sampling; however obsolete liquid rate may lead to incorrect oil production estimation.

Domain Knowledge Review

The machine learning was proposed on the Handil field to estimate liquid production rate on gas lifted oil wells, as well as to perform gas production rate correction on the water-out gas wells equipped with orifice flow meter. In general, all of required variables data for machine learning development are available. Below are the basic theories shall be keep in mind to ensure that machine learning process and its product are in line with the domain knowledge.

Estimating Liquid Production Rate from Temperature Profile Approach. The calculation of well's production rate using temperature profile approach is a method that adopts the temperature profile calculation on multiphase fluid flow inside conduit proposed by Ramey (1962), Shiu & Beggs (1980), Hasan & Kabir (1996) and Sagar et al. (1991), which was later implemented to predict liquid rate on gas field wells by Setiadi (2022). A typical temperature profile with respect to depth under shut-in and flowing condition is illustrated in Figure 7.a. The well's temperature profile under shut-in (static) conditions represents the gradient profile of geothermal, where temperature increases with depth according to the geothermal profile. However, under flowing conditions, the temperature profile for oil, gas, or multiphase flow depends on: produced fluid condition such as the phase fluid proportions and flow rates; and surrounding conditions of specific heat transfer (u-value), pipe diameter, and geothermal gradient.

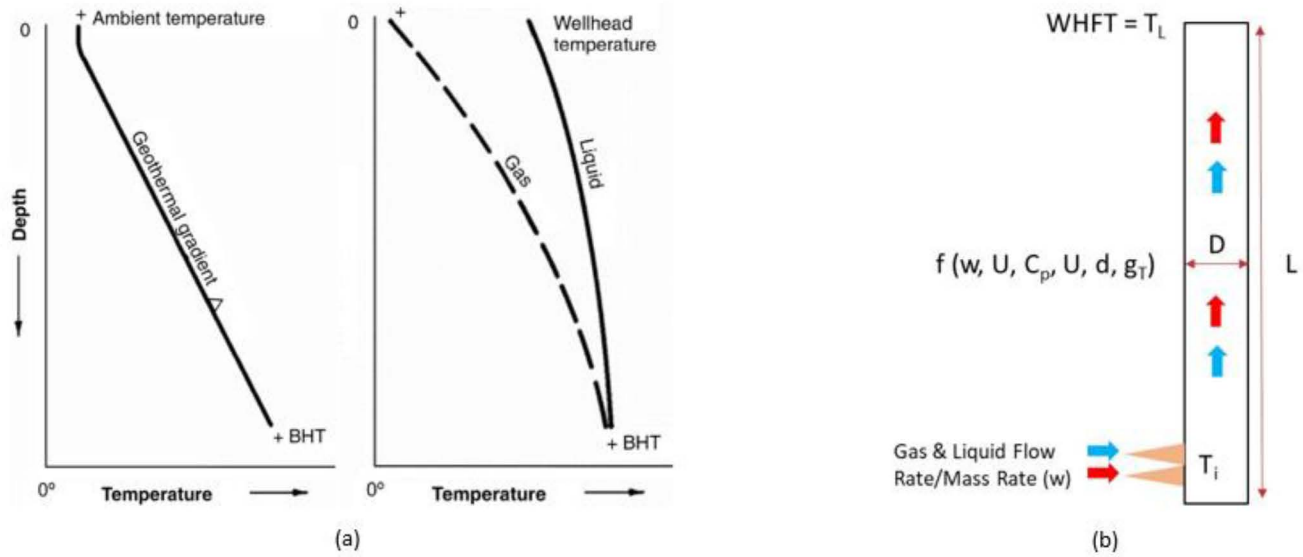


Figure 7—Temperature profile illustration (a) under shut in and flowing condition; (b) flow elements

More detail referring to figure 7(b), well head temperature on the flowing well could be calculated using equation 1 (Ramey, 1962):

$$T_L = T_i - g_T\left[L - A\left(1 - e^{(-\frac{L}{A})}\right)\right] \quad \text{(eq.1)}$$

Using above equation, the mass flow rate (part of A), combination of gas and liquid, could be reverse calculated or predicted by knowing the temperature at a certain point (T_L), the wellhead temperature

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(WHFT) in this case, and bottom hole temperature (T_i). To highlight, liquid production rate has strong relationship with gas production rate, surface temperature and bottom hole temperature.

Gas rate correction from orifice meter reading. One of production rates measurement approaches is the utilization of orifice flow meter. This device, placed in a pipeline, consists of a plate with a hole in the middle at a specific opening size. This hole creates a pressure drop at various flow rates, resulting in a differential pressure between the upstream and downstream of the plate.

In Handil field, particularly in most of well installations after 2015, an additional orifice flow meter is installed at flow line located downstream of the well choke. The measured differential pressure is converted into a gas rate reading (orifice gas rate), assuming full of gas condition even if liquid flow exists (multiphase flow).

Mathematically, the mass rate can be calculated by knowing the fluid properties, the ratio between the orifice hole and the pipe's internal diameter (beta ratio / β), and the differential pressure (Δp), as per equation 2 referring to ISO 5167-1:2003. It's important to note that this calculation is usually effective in single-phase flow (liquid or gas).

$$q_m = \frac{Cd}{\sqrt{1-\beta^4}} \epsilon \frac{\pi}{4} d^2 \sqrt{2\rho\Delta p}\,\mathrm{q} \qquad \text{(eq.2)}$$

Variable related to orifice plate then grouped and named as orifice coefficient rewritten in equation 3,

$$q_m = (Orifice_coef)\sqrt{2\rho\Delta p} \qquad \text{(eq.3)}$$

The real gas rate could be corrected referring to equation 3 when the liquid flow exists. The mass rate (q_m) and fluid density (ρ), which are combination of gas and liquid production rate, could be iterated to get minimum discrepancy with real differential pressure (Δp) at known orifice coefficient component. The iteration is easier when one of fluid component, real gas rate or liquid rate, is known. This basic concept emphasizes that the real gas production, orifice coefficient, differential pressure reading, and real liquid production have strong relationship.

Machine Learning Deployment for Liquid Rate Prediction

The whole process of supervised machine learning to predict liquid rate could be summarized on flow chart illustrated on figure 8.

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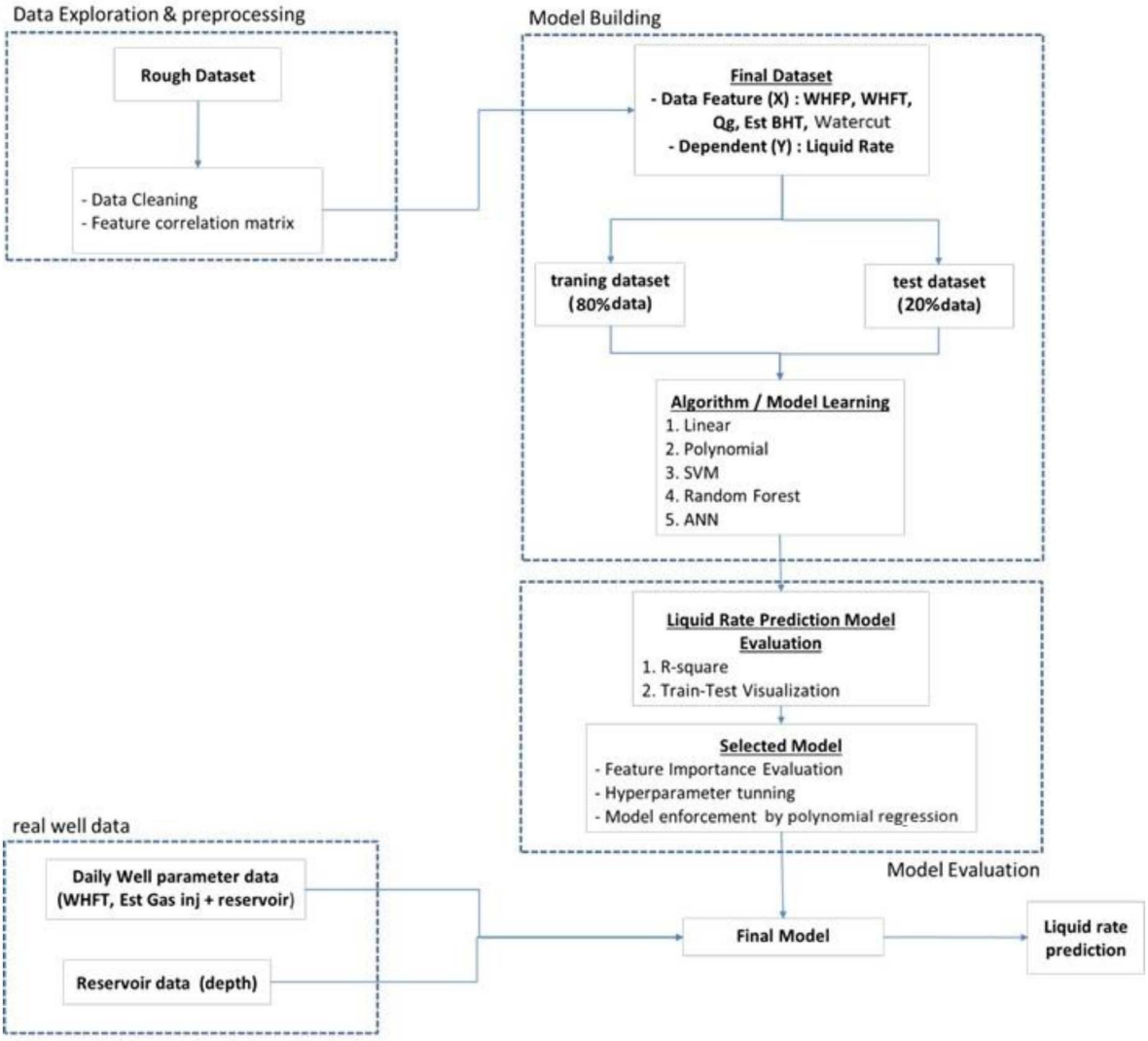


Figure 8—Flow chart of machine learning process to estimate well's liquid rate production

Data exploration and preprocessing. For the purpose of estimating the liquid rate based on wellhead parameter data, a total of rough 1743 well test data were collected. They were divided into input features (X) and dependent variable (Y). The input features (X) consist of wellhead flowing pressure (WHFP), wellhead flowing temperature (WHFT), combination gas injection and reservoir gas rate (Qg), watercut, and reservoir temperature (est BHT) represented by reservoir depth. The dependent variable (Y) represents the actual equivalent liquid rate (Qliq_eq).

Equivalent liquid rate (Qliq_eq) is basically conversion or liquid rate (blpd) into mass rate. It accommodates water and oil density factor to be more relevant with equation 1.

Data quality is the most important aspect in machine learning to produce accurate prediction. Data cleaning was performed to eliminate dual string data and commingled well production data, resulting in 1424 final data sets.

Dual string. Simultaneously active dual-string wells can lead to interference in WHFT parameters. For example, a dual-string well with string #1 produces 3000 barrels per day (bpd) and string #2 produces 500 bpd. In theory, the string with a higher liquid rate should result in a higher WHFT. However, because string #2 intersects with string #1 at the surface, the temperature of string #1 affects the WHFT of string #2. This can cause confusion in production data interpretation for dual-string wells. This such data samples shall be removed

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Comingle production. There are several wells whose flow paths on the surface are connected to other wells before entering the production test manifold header, as illustrated in Figure 9. Production test data from comingle wells provides the production flow rate for both wells, where the exact production allocation for each well is unknown unless one of the well is shut in. This comingle well test data is also excluded from the data samples.

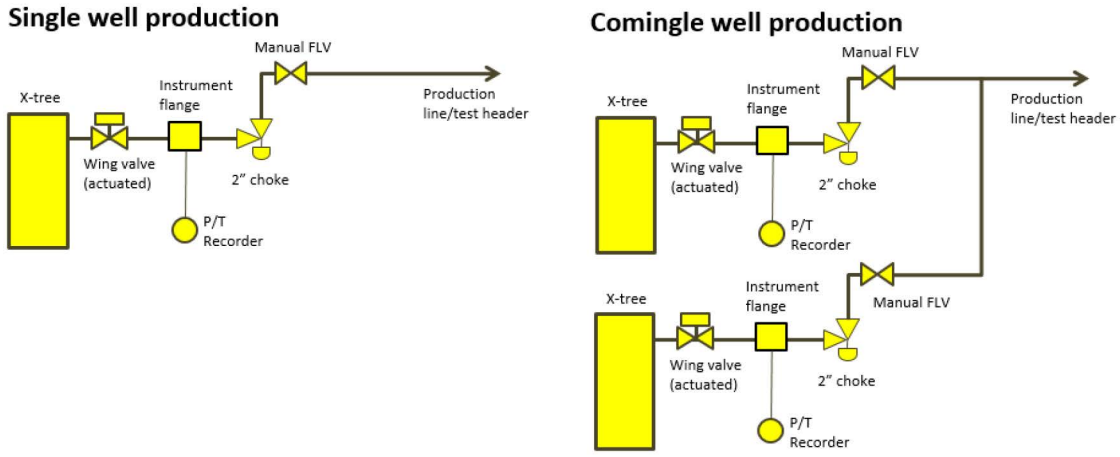


Figure 9—Single and comingle wells illustration

This final dataset was then analyzed for each parameter correlation using Spearman correlation. The objective was to gain an understanding of the relationships between variables. The correlation matrix is displayed in Figure 10. As the initial evaluation, all features were used for the analysis.

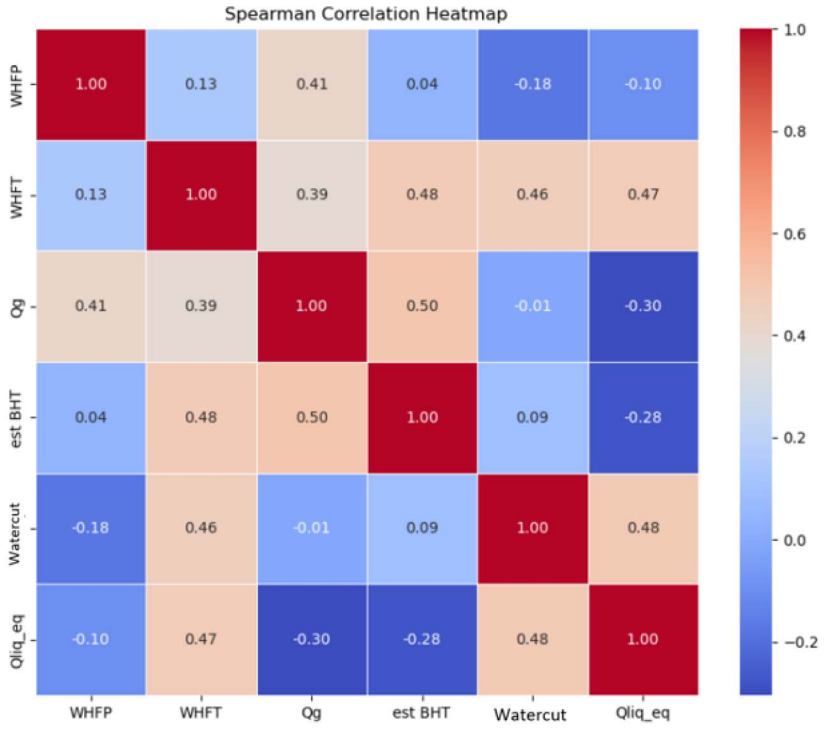


Figure 10—Heatmap correlation matrix

Model Building

There were two stages process in this phase, which were data splitting and algorithm evaluation.

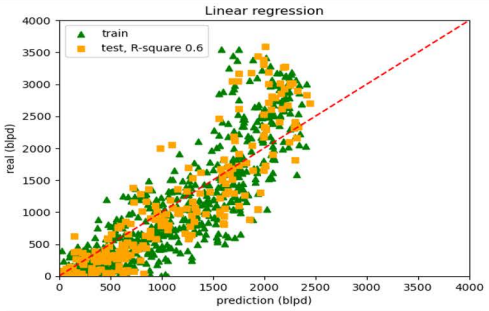
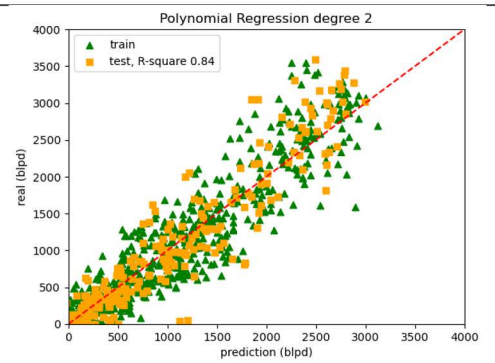
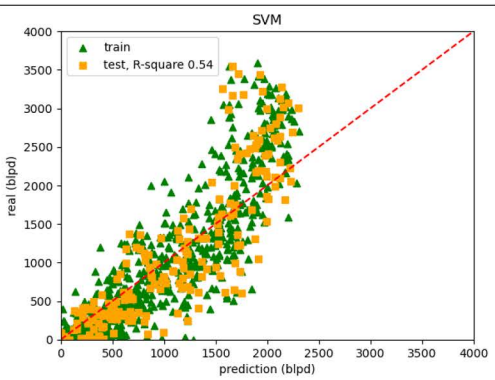
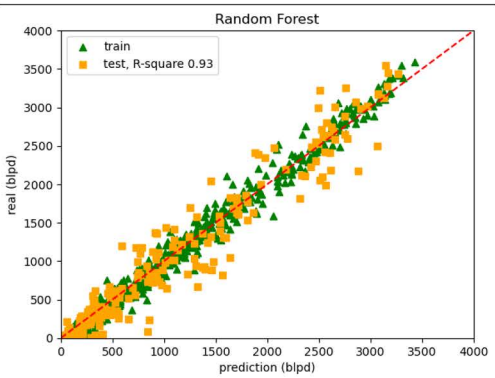
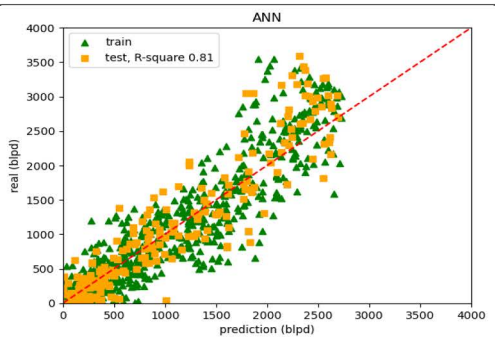
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Data splitting. The dataset was divided into two parts: the training set and the testing (or validation) set. The training set was used to train the model, while the testing set was used to assess its performance. The data split composition was set to 80:20%. At this stage, a random state was set to a specific number to ensure consistent estimation results.

Algorithm evaluation. Several algorithms /models were tested, including linear regression, polynomial regression, support vector machines (SVM), random forest regression, and artificial neural networks (ANN). In this stage, hyper-parameters were adjusted to obtain optimal evaluation results. Evaluation was represented not only by the calculated R-square value but also by comparison between estimated fluid and actual fluid to provide a better visualization of model accuracy. The results are illustrated in Table 1.

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Table 1—Model Performance Summary

Model	R-square (fraction)	Visualization
Linear	0.6	
Polynomial	0.84	
SVM	0.54	
Random Forest	0.93	
ANN	0.81	

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Model Evaluation

After the best model selected, the random forest, a re-evaluation was conducted to test and identify the input features having the most significant impact on predictions. This allows the possibility for the removal of less important features. The higher the number, the more important the feature for predictions. Simulation results show that the dominant features were wellhead flowing temperature (WHFT), and reservoir temperature (est BHT), while gas rate (Qg), watercut and WHFP cumulative importance was less than 0.2 as seen in the attached Figure 11. However, as these parameters could be updated daily, all of the feature were maintained for the prediction process.

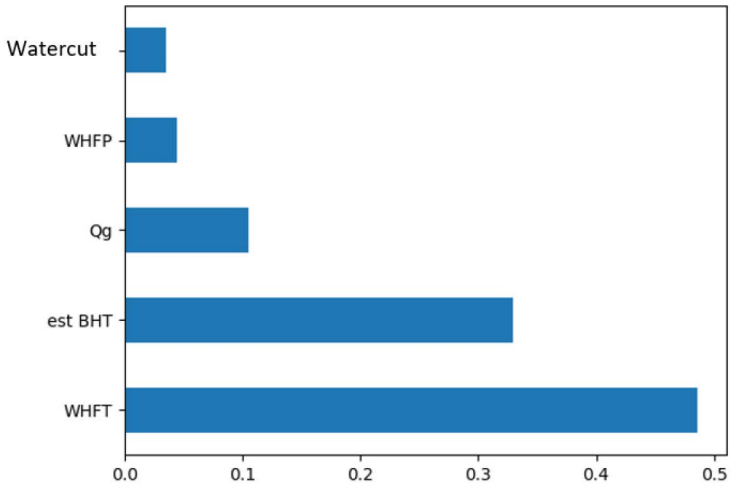


Figure 11—Feature importance evaluation

As per the figure above, the importance of the Qg, WHFP and watercut feature is only less than 0.1 each. It indicates that prediction is less sensitive to the actual gas flow rate, WHFP and watercut, which provides an advantage in prediction. In real cases, the Qg (gas flow rate) is typically the sum of measured gas injection rates and reservoir's associated gas (calculated GOR), usually around 0.1 to 0.5 MMscfd. The impact of an inaccurate GOR assumption would be insignificant to the prediction variation.

To measure prediction strength, testing was conducted on data that the model has never seen before (blind data). Random forest is known for its ability to provide high accuracy but is subject to overfitting and less sensitive on the data range outside the training set. Since WHFT was the most dominant parameter, WHFT sensitivity analysis was carried out while keeping depth (est BHT) and gas injection rate (Qg) and watercut at constant values. It was observed that the random forest prediction was flat outside the training data as illustrated on example case analysis at a fixed gas rate of 2 MMscfd and a depth of 2200 mTVD (est BHT estimation of 101.6 degrees Celcius). At above 90 degrees celcius, the prediction is flat and also some overfitting occurred at specific points, as illustrated by the blue line in Figure 12.

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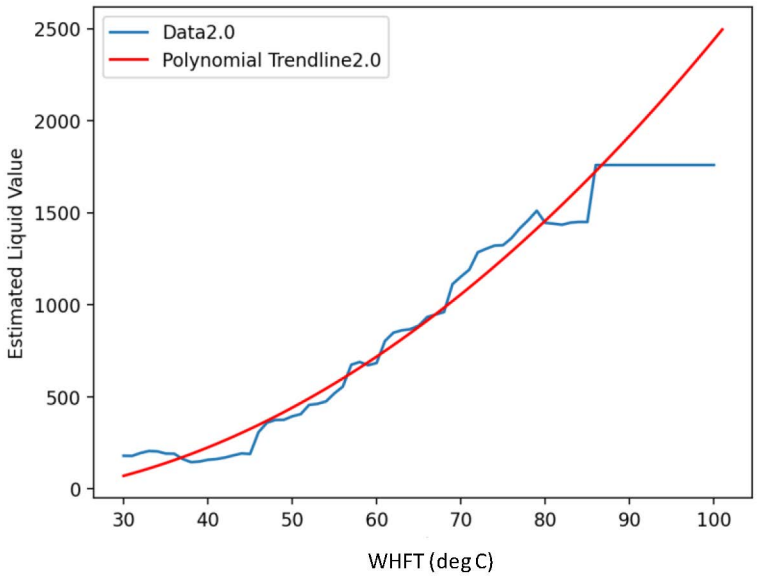


Figure 12—Example of predicted liquid rate based on WHFT sensitivity test

To address this issue, combination regression approach was employed by adding polynomial regression on sensitivity analysis results. This method provided more robust prediction on blind data in real-world applications. The additional polynomial regression is illustrated on the red line in figure 12. Using this approach, there will be 2 step regressions applied for the well prediction deployment. First, Random Forest regression is performed at range of WHFT value at fix depth (est BHT) and Qg. Second, Polynomial regression then being executed to get final liquid rate prediction based on the trend from the first step. The combination regression method was tested on the whole dataset and it still provided high accuracy prediction, the blue dot, at r-square of 0.91 as shown in figure 13.

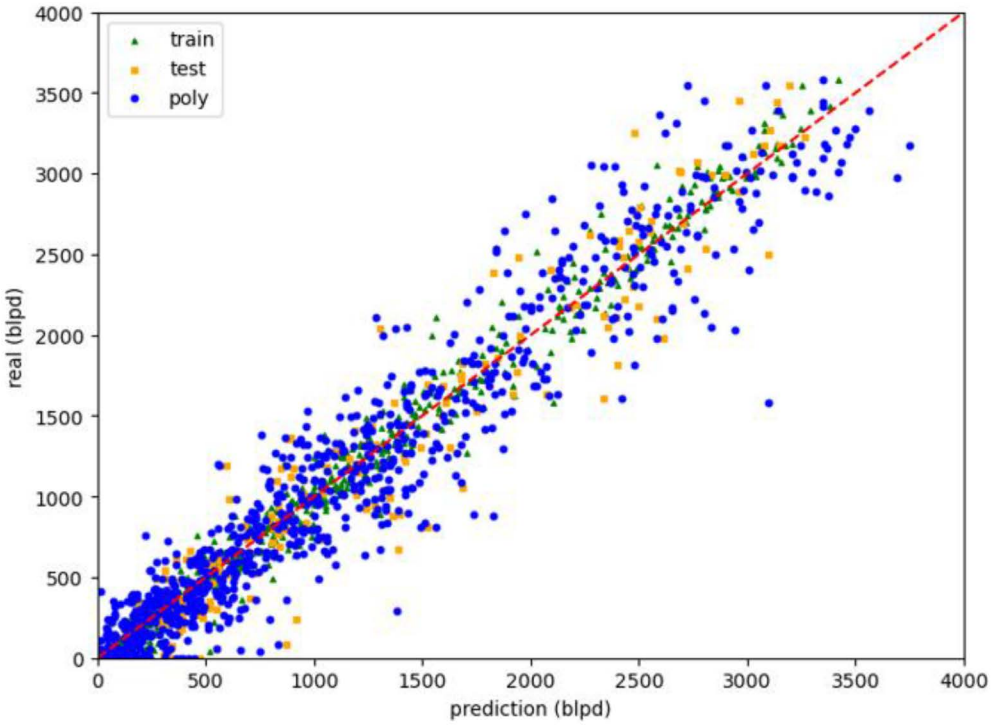


Figure 13—Model accuracy test by using combined regression method on all dataset

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Machine Learning Deployment for Gas Rate Correction on Water-out Gas Well

Under slightly different process, second machine learning was developed for this purpose. It started from data exploration until model evaluation. The process was divided into 2 stages. The first stage checked if the liquid production rate could be alternatively predicted by using orifice meter reading as dominant features (independent to previous method). Second stage corrected real gas rate on water-out gas well by using sensitivity method. The main highlighted points are summarized below:

1st Stage: Data exploration and preprocessing. 167 clean data of well's production test completed with orifice meter reading during the test were prepared. Input features (X) are delta_Qg which is the subtraction between calculated gas rate displayed on the orifice flow meter (Qg orifice) and real gas during production test (Qg well), differential pressure reading on the orifice (Δp), wellhead parameter data (WHFP/T) and orifice coefficient data (calculated beta ratio, coefficient of discharge, and orifice diameter refers to equation 2). The dependent variable (Y) represents the actual liquid rate (Qliq).

1st Stage: Model Building. The dataset was divided into two parts: the training set and the testing (or validation) set with composition ot 75:25%. Several algorithms were tested with best result on random forest regression. The prediction result comparison to actual rate (in BLPD unit) is illustrated on figure 14.

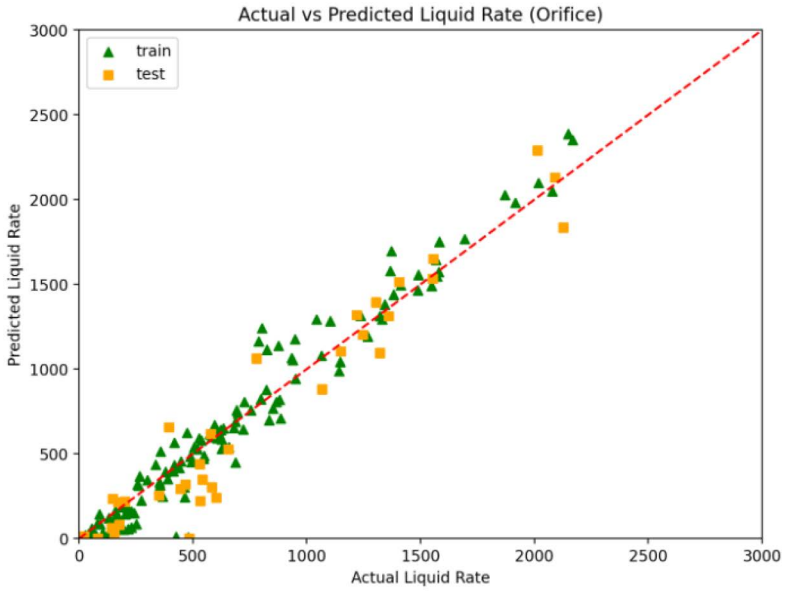


Figure 14—Actual and prediction rate comparison on random forest regression model

1st Stage: Model Evaluation. Feature's domination was checked on selected model. It showed that the importance of wellhead flowing temperature (WHFT) feature was insignificant. The most importance feature was the delta gas (delta_Qg). The feature importance evaluation result is shown on figure 15(a). By this new feature important map's figure, the 1st stage objective was achieved. There are now two independent liquid rate prediction methods, wellhead prediction controlled by WHFT as the most important feature and orifice prediction controlled by delta gas from orifice reading and real produced gas as the most important feature.

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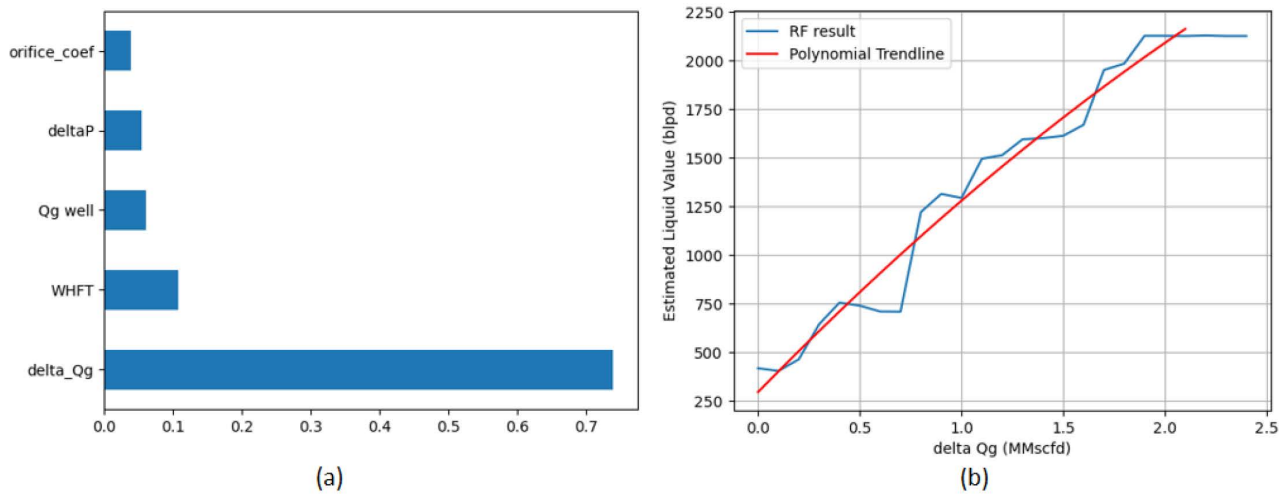


Figure 15—(a) feature importance evaluation result (b) delta Qg sensitivity calculation on specific fix feature value

Under similar process, additional polynomial regression was deployed to have strong liquid rate prediction result and minimum over-fitting on the blind data. The difference, sensitivity calculation was performed on delta Qg as highest importance feature. Figure 15(b) illustrate an example of sensitivity calculation performed on fix parameter input of 84 deg C WHFT, differential pressure (Δp) at 136 inH20, orifice coefficient at 0.001566923, and real well's gas (Qg well) at 0.9 MMscfd.

2nd Stage: Real gas rate correction. This is the trickiest part. There are now two independent machine learning methods to estimate liquid rate. Thus, the liquid rate could be first predicted by using well head temperature, est BHT, gas rate and watercut with first assumption that the orifice meter reading is valid.

For example, let assume that liquid estimation based on first method prediction is 2000 blpd. A combination of random forest and polynomial regression was performed multiple time using 2nd prediction method at various variation of actual gas (Qg well) and delta gas (delta Qg) feature value. For example, if the orifice meter reading is 2.8 MMscfd, the liquid rates are predicted and plotted starting from combination of Qg well 0.1 MMscfd and delta Qg 2.7 MMscfd, until Qg well 2.7 MMscfd and delta Qg 0.1 MMscfd. The plot is illustrated on figure 16 below. The corrected gas rate is the crossing line between the blue line (sensitivity estimated liquid rate) and the red straight line of 2000 blpd liquid predicted before.

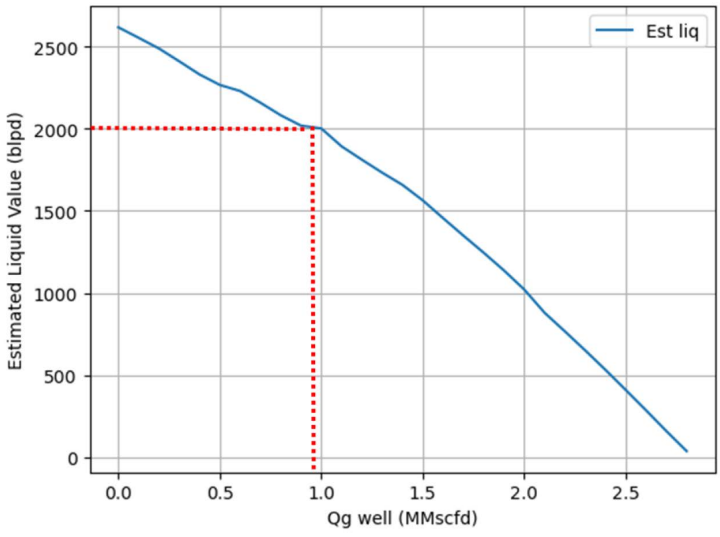


Figure 16—Example case finding corrected gas rate value

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Result and Limitation

Better field's produced liquid figure

The liquid rate estimation was applied on all producing wells in Handil field. The result, newly predicted liquid has better accuracy (~90%) compared to previous method referring only to latest production test activity (~80%) as illustrated on figure 17.

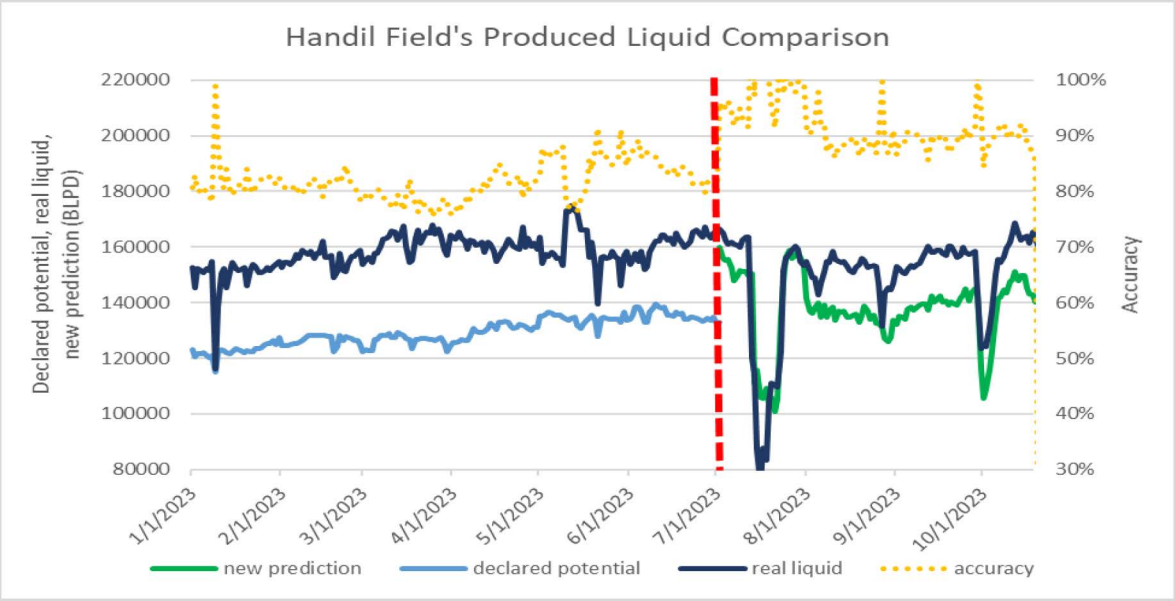


Figure 17—Handil filed's liquid production comparison

Incline decline identification

Using the similar case history for comparison purpose, H-Q-X well, the dynamic liquid rate prediction could be observed when well head parameter changed as seen on figure 18. It provided additional engineer insight for well optimization respecting the production limitation on specific well condition (e.g well completed with chemical sand control).

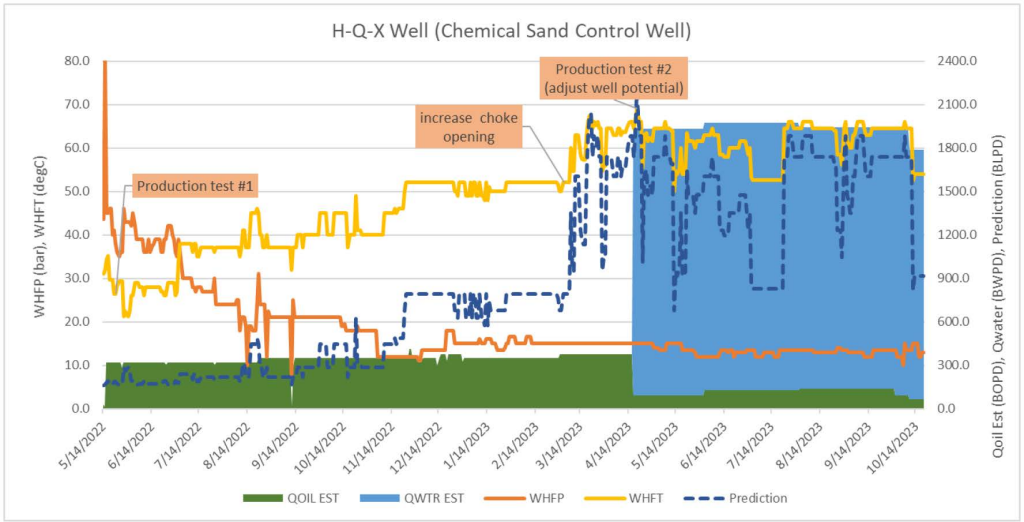


Figure 18—Chemical sand control well's liquid production evolution by using new prediction method

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There was another case where well head parameter changed and interpreted as liquid production reduction. Necessary recovery action by increasing choke opening were performed to safely bring back the well to previous well potential. The explained example case, H-XA-X well, is shown on figure 19.

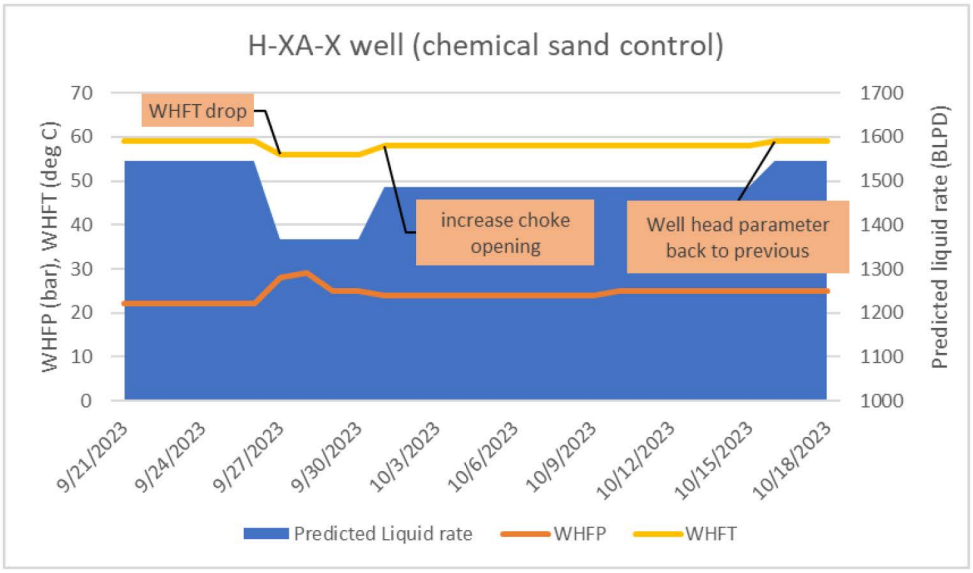


Figure 19—Example well production optimization based on estimated liquid rate

Evaluation was carried out on the sand production event mentioned earlier in H-X-X well case. Using the newly liquid estimation approach, well's flow rate was up to 1200 BLPD illustrated on figure 20. This value was much higher compared to declared well's liquid production potential which is only around 450 BLPD. This new method revealed new information that production rate higher than 1000 BLPD (red line on figure 20) may cause sand event on the well installed with specific thru tubing screen type. This case sample can be used for future production surveillance reference to avoid sand event which impacts on well potential loss.

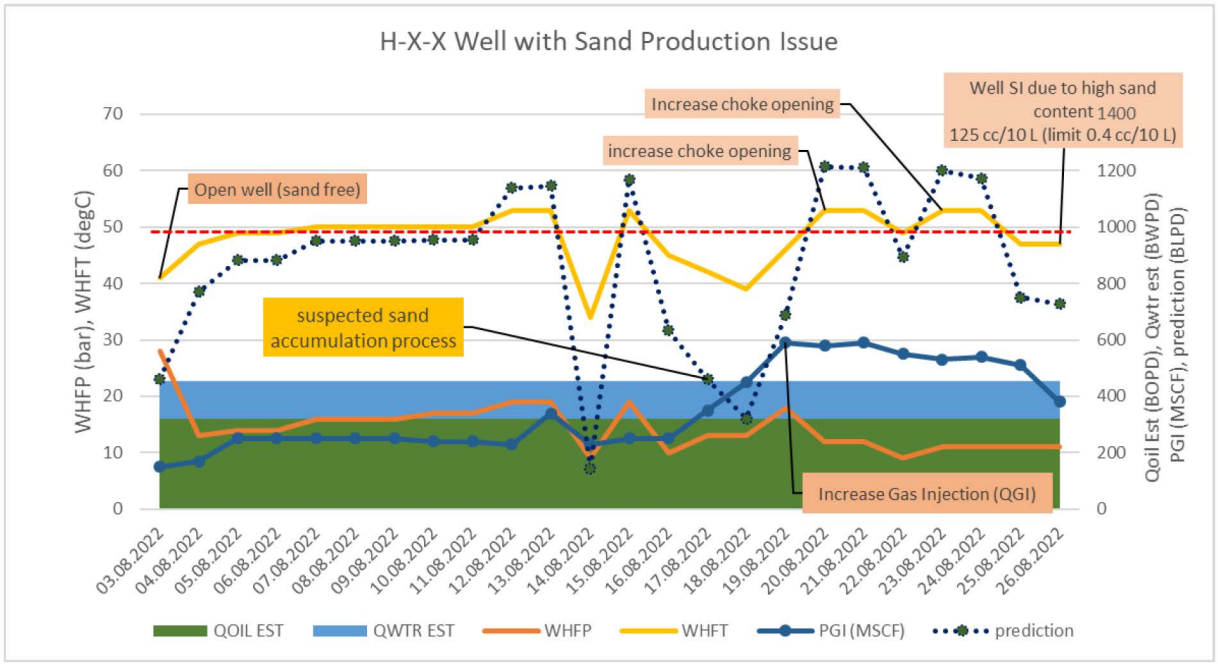


Figure 20—Example Sand production event evaluation

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Gas production rate correction on water-out gas well

H-V-X is one of water-out gas well example. This well was equipped with additional orifice flow meter which parameter was daily updated by operator visit. There was a gas rate limitation on the flow line at 5 MMscfd, thus the rate was maintained at 5 MMscfd referring to orifice meter. However, there was a doubt that the real gas was actually lower than 5 MM because an indication of liquid flow from the sampling point. Later, a mobile testing unit was mobilized and confirmed that the actual gas rate was 4.5 MMscfd with 650 blpd liquid rate. Using the developed method as illustrated on figure 21, actual rate prediction was 4.42 MMscfd with 540 blpd liquid, which quite match with testing barge result. It was later used as actual gas rate reference.

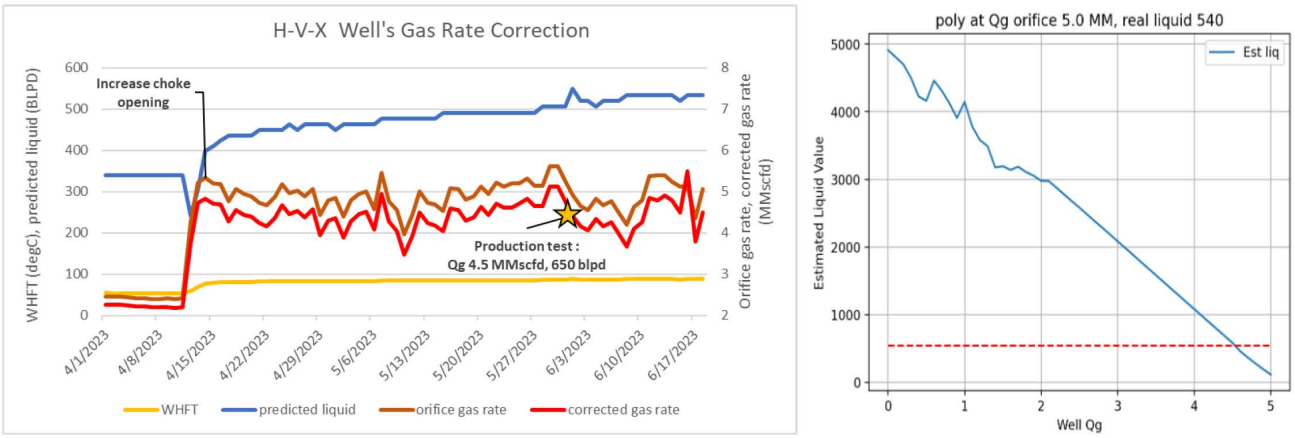


Figure 21—Example of gas well rate correction from orifice reading

Limitation

Though the prediction result is quite impressive, this developed method still has some limitation:

Mingle reservoir production. The machine learning was trained with assumption of single depth reservoir production as one of feature. In real well application, In several gravel pack wells and main zone, two or more reservoir are produced mingle creating difficulties to define the reservoir depth feature (est BHT). This condition could be mitigated by individual well review to evaluate which dominant reservoir contributes based on well history. However, direct depth averaging could be selected as alternative if no clue at all on which contributing reservoir. The prediction will be slightly inaccurate but still acceptable due to depth of reservoir feature dominance is lower than well head flowing temperature.

Active dual string well. As mentioned earlier, active dual string well will have risk of temperature interference between each string which lead to inaccurate prediction on one of the string.

Conclusion

The successful deployment of supervised machine learning in the Handil oil field has addressed monitoring challenges arising from the absence of well-level metering and sedimentation limitations. Two machine learning models were specifically designed to estimate liquid and correct gas rates by leveraging available parameters. Among tested algorithm, the random forest regression emerged as the most accurate, achieving an impressive accuracy rate of 93%.

The impact of these machine learning models on daily liquid production accuracy elevated the precision from 80% to an impressive 90%. Moreover, these models demonstrated their capability to detect production changes, enhancing the field's adaptability to dynamic conditions. Additionally, it offered insights by providing valuable information for optimizing sand control well's production.

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One of the key strengths of this approach lies in its potential for replication on fields with limited monitoring data. Overall, the integration of supervised machine learning has proven to be a transformative solution, significantly improving well surveillance and providing valuable insights in the challenging environment of the Handil oil field.

Nomenclature

Ti	temperature at fluid entry (L=0)
TL	temperature at location L
gT	geothermal gradient
A	relaxation distance, wCp/πdU
w	mass flow rate
Cp	specific heat of the flowing fluid
d	pipe diameter
u	overall heat transfer coefficient
L	distance from fluid entry
Qm	mass rate through orifice (kg/s)
Cd	coefficient of discharge (v/v)
β	beta ratio (d/D)
d	orifice diameter (m)
D	pipe internal diameter (m)
ρ	fluid density upstream (kg/m3)
Δp	pressure drop (Pa)

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